

EMCH380 – Project Management Network Generation

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Slides are developed from instructor's notes and other references.

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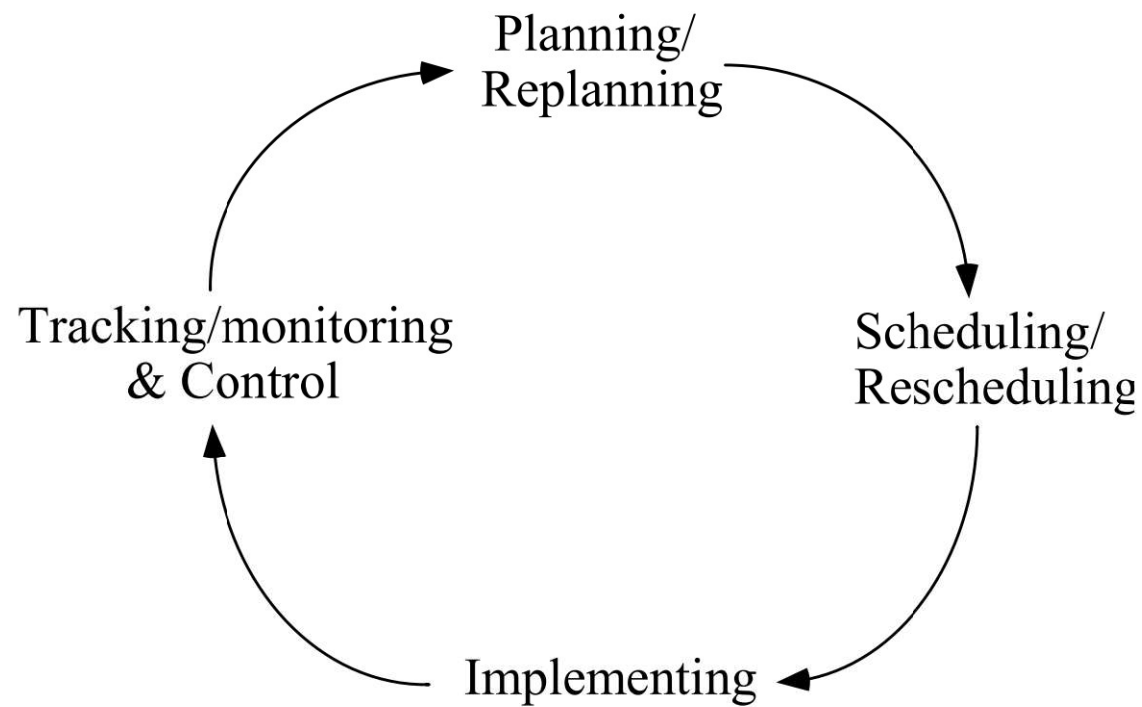
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2. Activity Relationship / Precedence Relationship
 - Finish to Start (FS)
 - Start to Start (SS)
 - Finish to Finish (FF)
 - Start to Finish (SF)
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5. Duration Estimation & Resource Assignment

Introduction

Network Generation

DETAILED PLANNING & MONITORING

I



NETWORK GENERATION

- Identify Elementary Items of Work —Activities
- Establish Precedence Between Activities
- Prepare Graphical Display of Planning Info in the Form of a Network

EXAMPLE

- **Preliminary Activity List for Remodeling Chemical Laboratory:**

- Strip Room
- Repair Walls and Ceilings
- Repair Floor
- Lay Vinyl Floor
- Rough –in Plumbing and Electrical
- Finish Plumbing and

Electrical

- Replace Existing Fume Duct
- Install New Fume Hood
- Install Base Cabinets
- Install Wall Cabinets
- Install Chemical Sinks
- Paint Cabinets
- Paint Walls and Ceiling

- Obtain Vinyl Floor Covering
- Obtain Cabinets
- Obtain Fume Hood
- Obtain Chemical Sink
- Painter Availability

Activity Relationship / Precedence Relationship

Network Generation

ACTIVITY RELATIONSHIPS / PRECEDENCE RELATIONSHIPS

- Independent Activities
- Dependent Activities
 - Finish to Start (FS)
 - Start to Start (SS)
 - Finish to Finish (FF)
 - Start to Finish (SF)
- Other Constraints on Start and Finish Dates of Activities

FINISH-TO-START RELATIONSHIPS

- **Without Lag**

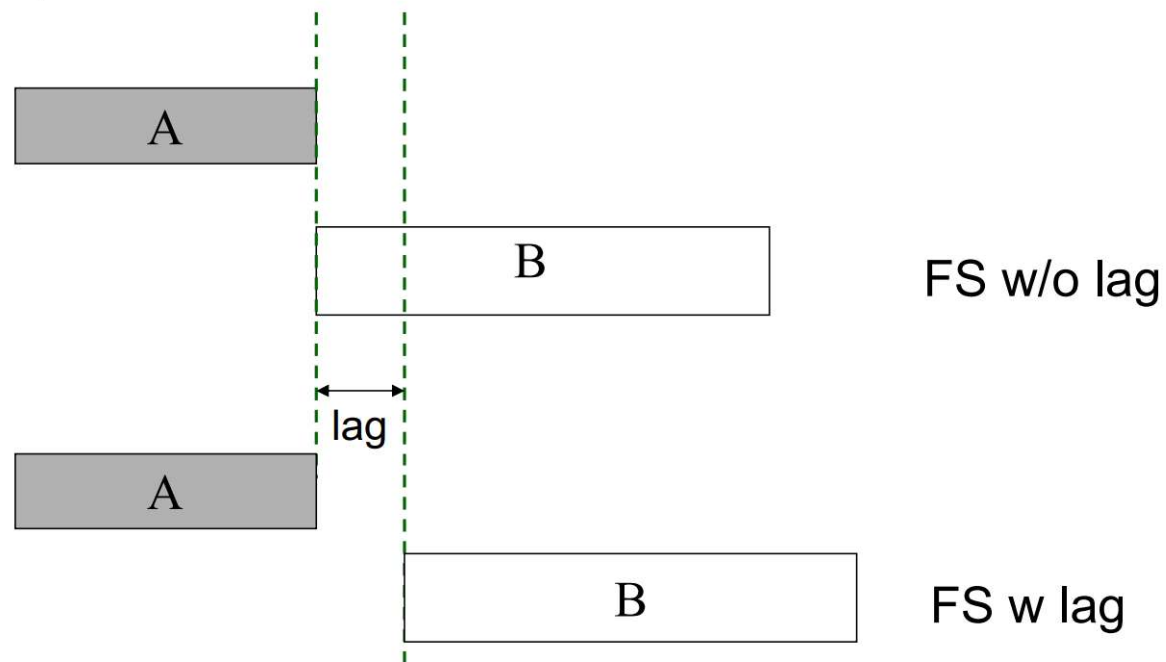
- Ex: Activity B has an FS relationship with Activity A
- Activity B cannot start before A is finished

- **With Lag**

- Ex: Activity B has an FS relationship with Activity A with a Lag of 2 days
- Activity B cannot start before 2 days after A is finished.

- *A lag is the amount of time added after finishing activity A before starting activity B. (ex: waiting for concrete to dry after pouring, before starting other dependent activities)*

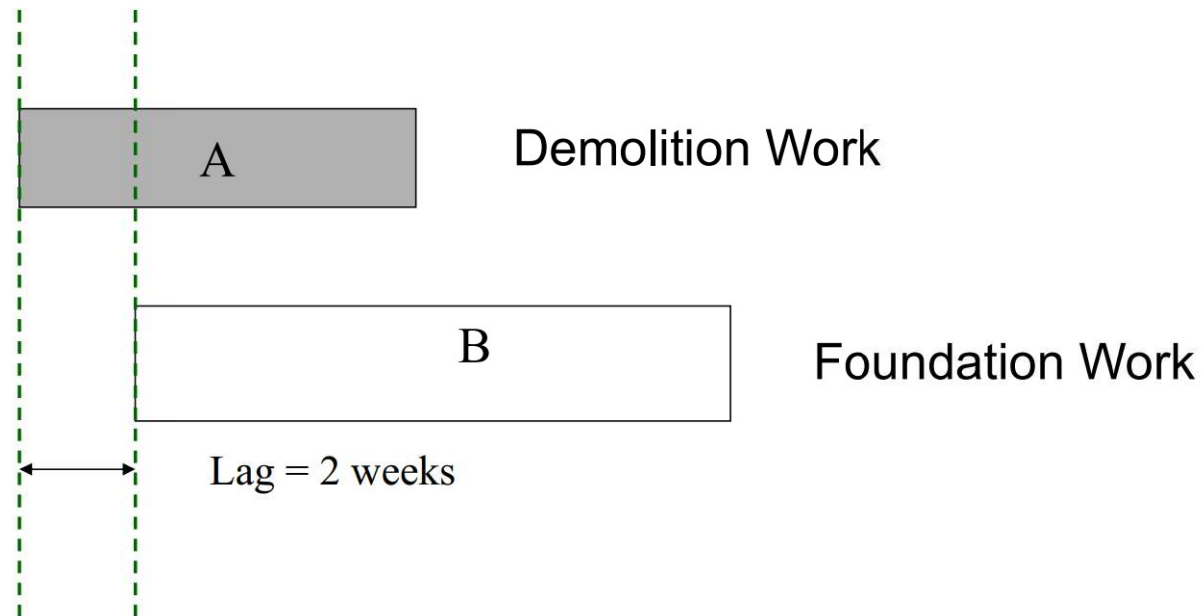
FINISH-TO-START RELATIONSHIPS (CONT'D)



START-TO-START RELATIONSHIPS

- Activity B has an SS relationship with Activity A with lag = 3 days
- Activity B can start 3 days after starting Activity A
- Ex: Foundation work (B) can start 2 weeks after starting the Demolition work (A).
- *Activity B cannot start unless activity A has started and a certain amount of time (the lag) has passed.*

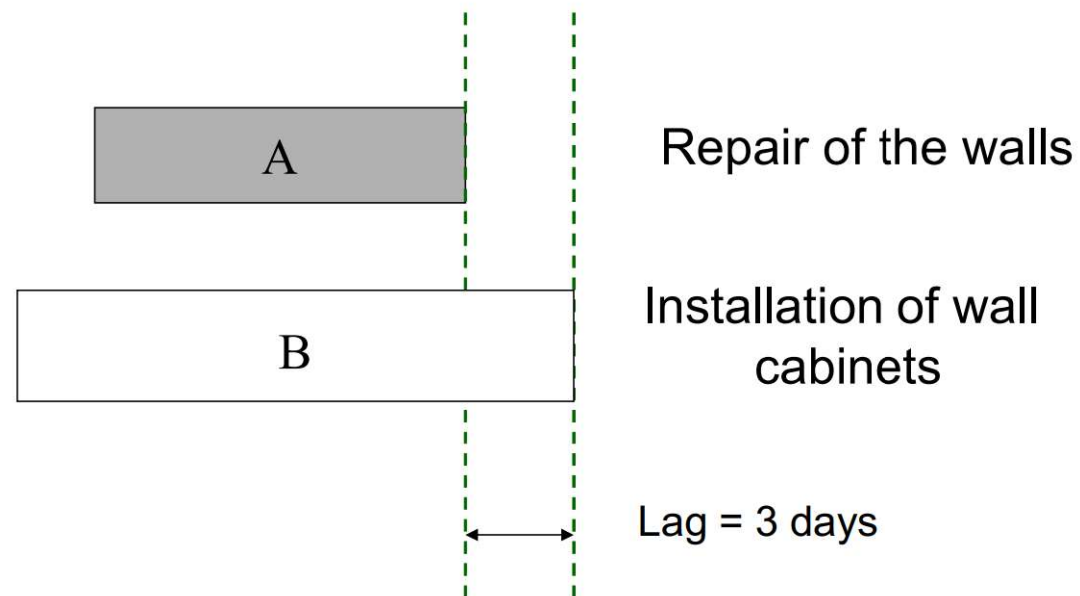
START-TO-START RELATIONSHIPS (CONT'D)



FINISH-TO-FINISH RELATIONSHIPS

- Activity B has an FF relationship with Activity A with lag = 3 days
- Activity B cannot finish before 3 days after finishing Activity A
- Ex: Installation of the Wall Cabinets (B) needs 3 weeks to complete after the Repair of the Walls activity (A) is completed (The installation of the wall cabinets will finish 3 weeks after the finish of the repair wall activities).

FINISH-TO-FINISH RELATIONSHIPS (CONT'D)



START-TO-FINISH RELATIONSHIPS

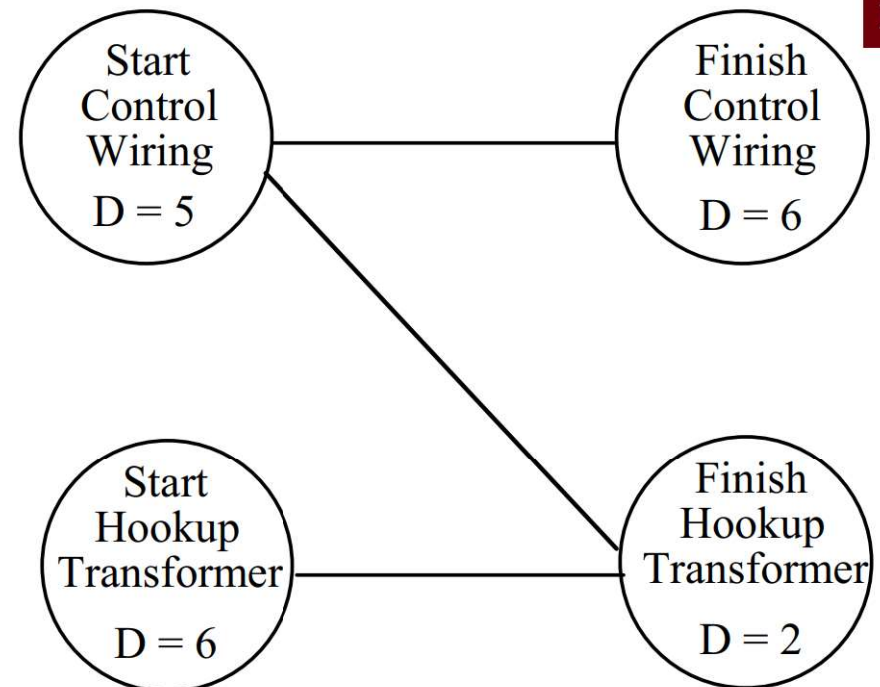
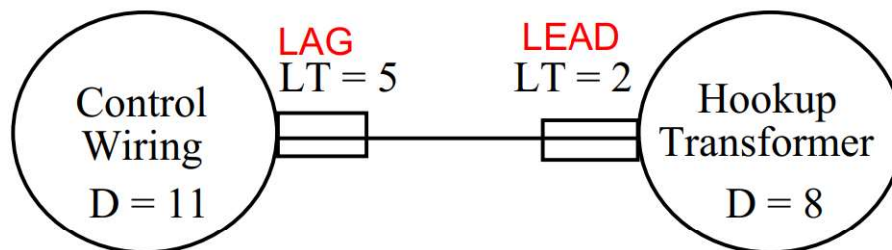
- Activity B has an SF relationship with Activity A with lag = 5 and lead = 2 days
- Activity B will finish 7 days after Activity A starts
- Activity B needs 2 days to finish after Activity A has been underway for 5 days

START-TO-FINISH EXAMPLE

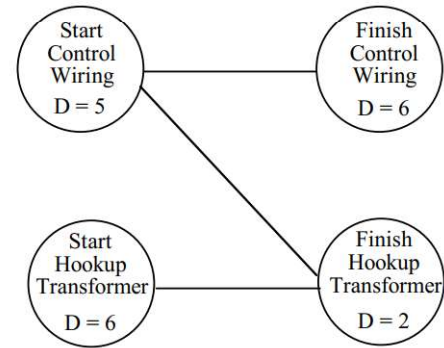
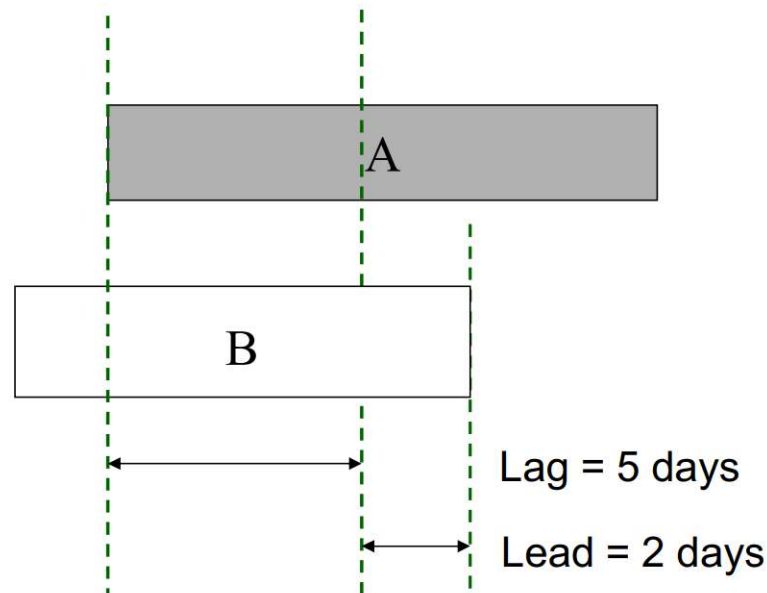
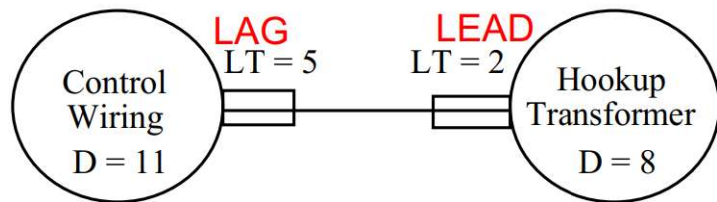
- **Scenario:**

- A large transformer is to be assembled and hooked up to a control wiring system.
- It is estimated that 5 weeks will elapse before the wiring is far enough to allow actual hookup of transformer to start and there must be 2 weeks allowed to complete the hookup operation
- 5 days after starting the control wiring system, the actual hookup of the transformer can be finished and finishing it will require 2 days, hence the above relationship.

START-TO-FINISH EXAMPLE (CONT'D)



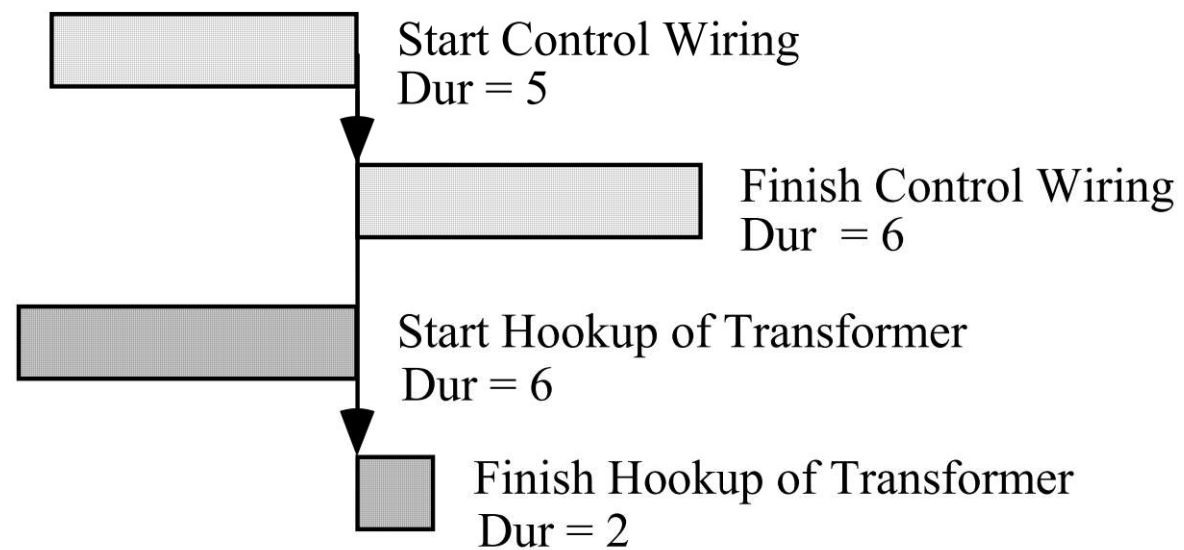
START-TO-FINISH EXAMPLE (CONT'D)



START-TO-FINISH EXAMPLE (CONT'D)



II



ACTIVITY LIST WITH DEPENDENCIES

No	Activity	Predecessors
1	Strip Room	
2	Repair Walls and Ceilings	1,5,7
3	Repair Floor	1,5
4	Lay Vinyl Floor	3,12,13,14
5	Rough -in Plumbing and Electrical	1
6	Finish Plumbing and Electrical	2,3,5,9,10,11,19
7	Replace Existing Fume Duct	1
8	Install New Fume Hood	2,3,16
9	Install 1/3 Base Cabinets	2,3,8,15
10	Install Wall Cabinets	2,3,7,15
11	Install Chemical Sink	2,3,5,9,17
12	Paint Cabinets	6,8,9,10,11,18
13	Paint Walls and Ceiling	2,3,6,8,9,10,18
14	Obtain Vinyl Floor Covering	
15	Obtain Cabinets	
16	Obtain Fume Hood	
17	Obtain Chemical Sink	
18	Painter Availability	
19	Install 2/3 Base Cabinets	2,3,9,15

ACTIVITY LIST WITH DEPENDENCIES

WBS	ID	Activity Description	Predecessor
1.1	10	Demolition	
1.2	15	Foundations	10FS
1.3	20	Underground Services	10SS+1d
2.1	25	Exterior Walls	40FS
2.2	30	Interior Walls	25SS+3d, 35FS
3.1	35	Roof Steel	40FS, 25FS
4.1	40	Floor Slab	15FS
5.1	45	Floor Finish	40FS, 70FS, 20FS, 25FS
5.2	50	Ceiling	35FS, 70FS, 80FS
5.3	55	Display Windows	25FS, 90FS
5.4	60	Paint	30FS, 55FS, 85FS, 75FS, 95FF+1d, 55FS
5.5	65	Roof finish	35FS
6.1	70	Rough Plumbing and heating	40FF + 1d
6.2	75	Finish Plumbing and heating	70FS
7.1	80	Rough electrical	40SS+1d
7.2	85	Finish Electrical	80FS
8.1	90	Rough Carpentry	25FS
8.2	95	Finish Carpentry	90FS, 55FS

Network Representations – Activity on Node (AON)

Network Generation

NETWORK REPRESENTATIONS

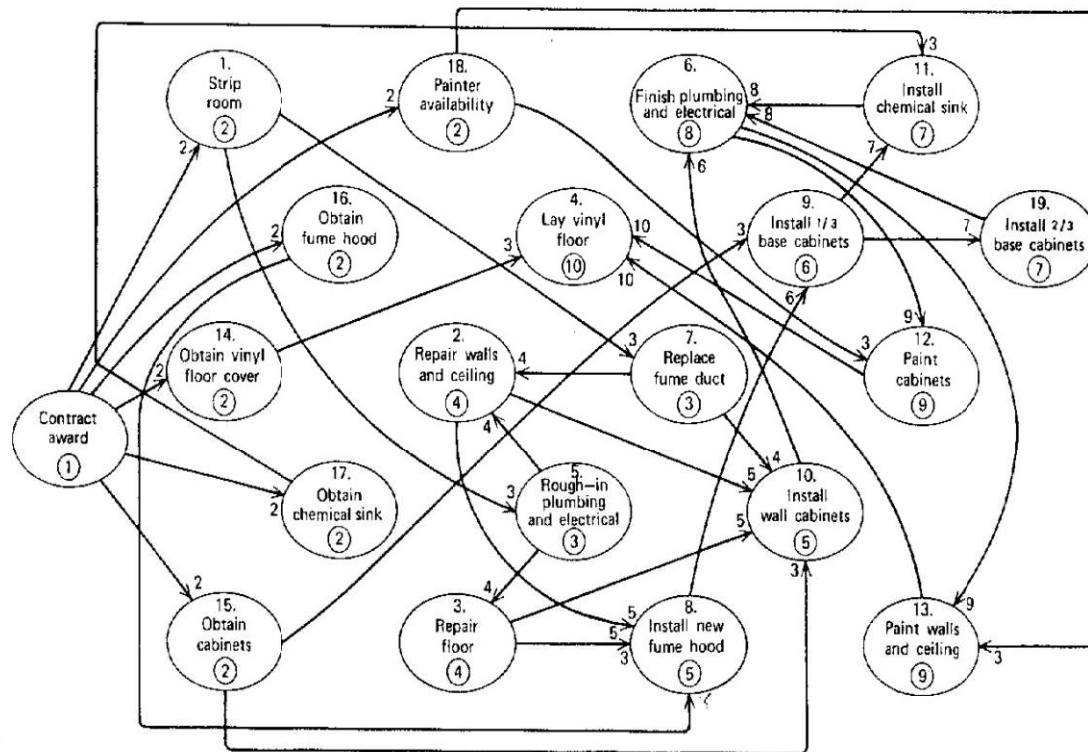
- Activity on Arrow Diagram (AOA)
- Precedence Diagram — Activity on Node Diagram (AON)

ACTIVITY LIST WITH DEPENDENCIES

No	Activity	Predecessors
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18	Painter Availability	
19	Install 2/3 Base Cabinets	2,3,9,15

AON – NETWORK EXAMPLE – BALL OF STRING DIAGRAM

III



ACTIVITY LIST WITHOUT REDUNDANCIES

- Each project activity is considered in turn and an activity number from its dependency list is selected.
- If the dependency list for this selected number contains a dependent activity that is the same as one of the activities in the dependency list of the original activity, then that number is removed from the original activity's dependency list.

No	Activity	Predecessors
1	Strip Room	
2	Repair Walls and Ceilings	1,5,7
3	Repair Floor	1,5
4	Lay Vinyl Floor	3,12,13,14
5	Rough -in Plumbing and Electrical	1
6	Finish Plumbing and Electrical	2,3,5,9,10,11,19
7	Replace Existing Fume Duct	1
8	Install New Fume Hood	2,3,16
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10	Install Wall Cabinets	2,3,7,15
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13	Paint Walls and Ceiling	2,3,6,8,9,10,18
14	Obtain Vinyl Floor Covering	
15	Obtain Cabinets	
16	Obtain Fume Hood	
17	Obtain Chemical Sink	
18	Painter Availability	
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3	Repair Floor	1 ,5
4	Lay Vinyl Floor	3 ,12,13,14
5	Rough -in Plumbing and Electrical	1
6	Finish Plumbing and Electrical	2,3 ,5,9,10,11,19
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10	Install Wall Cabinets	2,3,7,15
11	Install Chemical Sink	2,3 ,5,9,17
12	Paint Cabinets	6,8, 9 ,10,11,18
13	Paint Walls and Ceiling	2,3 ,6,8, 9 ,10,18
14	Obtain Vinyl Floor Covering	—
15	Obtain Cabinets	—
16	Obtain Fume Hood	—
17	Obtain Chemical Sink	—
18	Painter Availability	—
19	Install 2/3 Base Cabinets	2,3 ,9,15

BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS

- Remove Redundant precedence relationships
- Order activities in sequence steps
- Let sequence step order of activity i be S_i
- Initially $S_i=1$ for all activities with no predecessors and $S_i=\infty$ for all other activities
- Iterate over the set of all activities by computing:
 - $S_i = \max_{k \in p_i} \{S_k\} + 1$ until all activities have been assigned to a sequence step order

BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS - EXAMPLE

III

Activity	Depends upon
A	–
B	A
C	E, F
D	E
E	A
F	–

BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS - EXAMPLE

III

Activity	Depends upon
A	–
B	A
C	E, F
D	E
E	A
F	–

Activity	Depends upon
A	≠ ST
B	A
C	E, F
D	E
E	A
F	≠ ST
ST	–
FIN	B, C, D

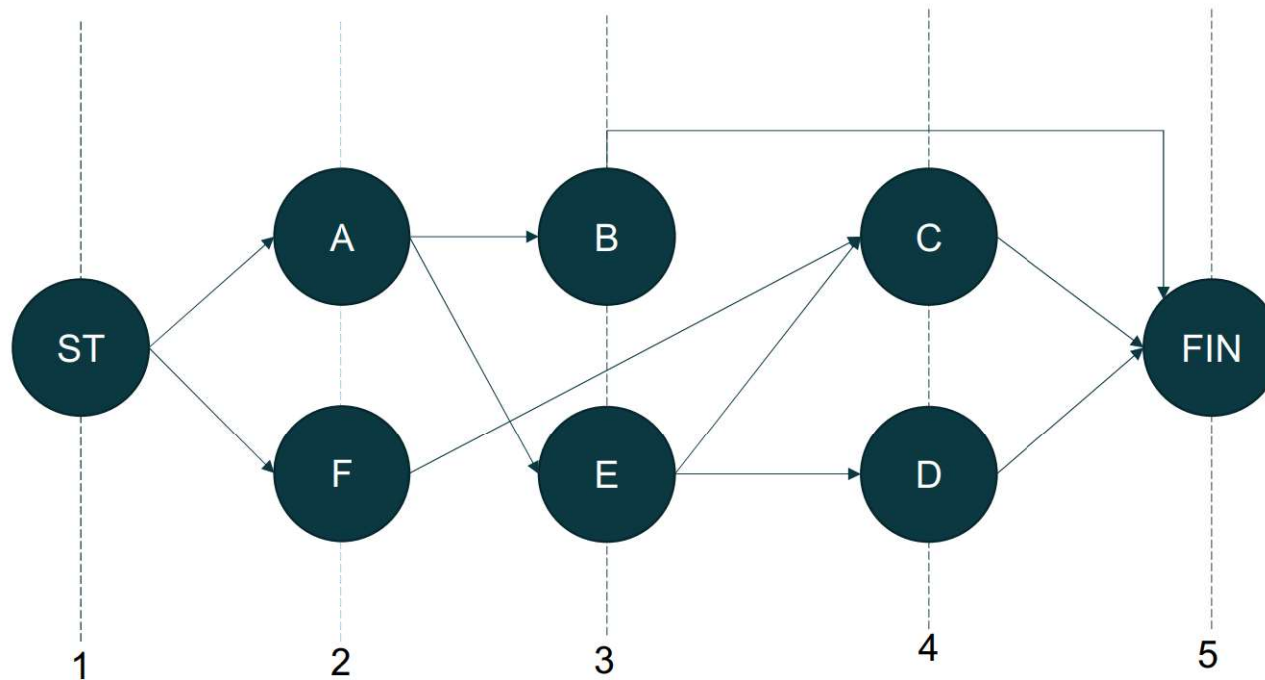
BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS - EXAMPLE

Activity	Depends upon
A	–
B	A
C	E, F
D	E
E	A
F	–

Activity	Depends upon
A	≠ ST
B	A
C	E, F
D	E
E	A
F	≠ ST
ST	–
FIN	B, C, D

Activity	Sequence Steps	
	Cycle 1	Cycle 2
ST	1	
A	2	
B	3	
C	≠	4
D	≠	4
E	3	
F	2	
FIN	4	5

BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS - EXAMPLE



BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS – ORIGINAL EXAMPLE

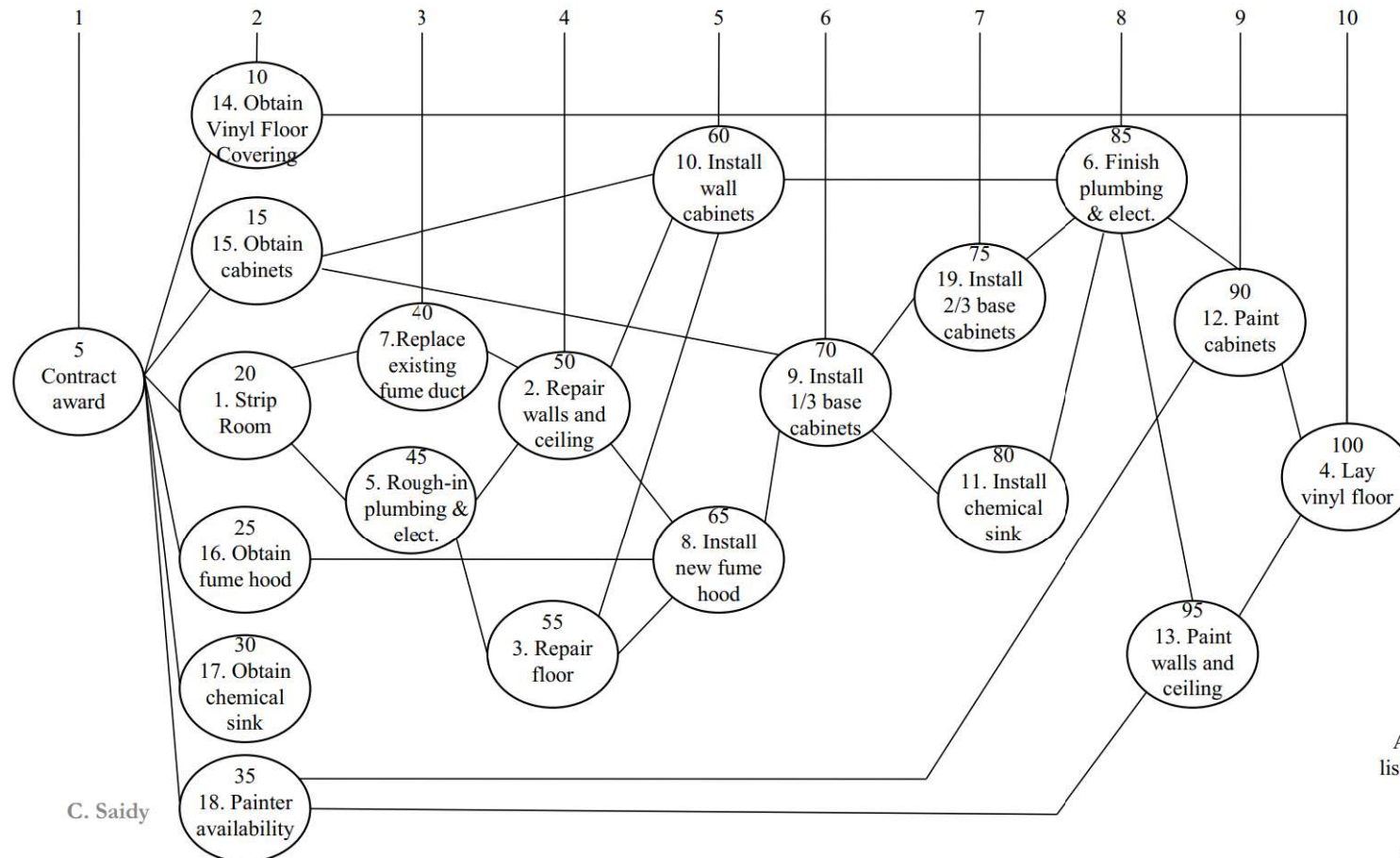
No	Activity	Predecessors	Sequence Steps								
			Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
1	Strip Room	-									
2	Repair Walls and Ceilings	5,7									
3	Repair Floor	5									
4	Lay Vinyl Floor	12,13,14									
5	Rough -in Plumbing and Electrical	1									
6	Finish Plumbing and Electrical	10,11,19									
7	Replace Existing Fume Duct	1									
8	Install New Fume Hood	2,3,16									
9	Install 1/3 Base Cabinets	8,15									
10	Install Wall Cabinets	2,3,15									
11	Install Chemical Sink	9,17									
12	Paint Cabinets	6,18									
13	Paint Walls and Ceiling	18									
14	Obtain Vinyl Floor Covering	-									
15	Obtain Cabinets	-									
16	Obtain Fume Hood	-									
17	Obtain Chemical Sink	-									
18	Painter Availability	-									
19	Install 2/3 Base Cabinets	9									

BUILDING PRECEDENCE DIAGRAM USING SEQUENCE STEPS – ORIGINAL EXAMPLE

No	Activity	Predecessors	Sequence Steps								
			Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
1	Strip Room	-	1								
2	Repair Walls and Ceilings	5,7			3						
3	Repair Floor	5			3						
4	Lay Vinyl Floor	12,13,14									9
5	Rough -in Plumbing and Electrical	1		2							
6	Finish Plumbing and Electrical	10,11,19							7		
7	Replace Existing Fume Duct	1		2							
8	Install New Fume Hood	2,3,16				4					
9	Install 1/3 Base Cabinets	8,15					5				
10	Install Wall Cabinets	2,3,15				4					
11	Install Chemical Sink	9,17						6			
12	Paint Cabinets	6,18								8	
13	Paint Walls and Ceiling	18								8	
14	Obtain Vinyl Floor Covering	-	1								
15	Obtain Cabinets	-	1								
16	Obtain Fume Hood	-	1								
17	Obtain Chemical Sink	-	1								
18	Painter Availability	-	1								
19	Install 2/3 Base Cabinets	9						6			

III

REMODELING CHEMICAL LAB EXAMPLE – PRESENCE DIAGRAM



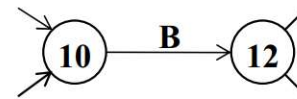
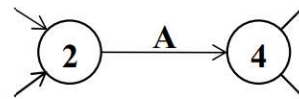
Network Representations – Activity on Arrow (AOA)

Network Generation

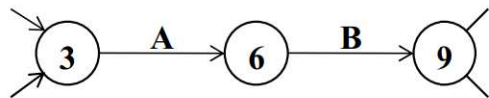
NETWORK REPRESENTATIONS

- Precedence Diagram — Activity on Node Diagram (AON)
- Activity on Arrow Diagram (AOA)

ACTIVITY ON ARROW – BASIC LOGIC PATTERNS

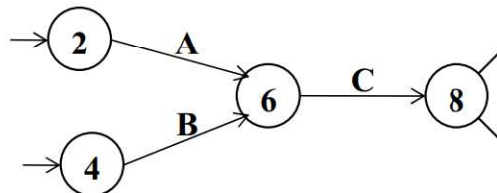


(a) Independent activities



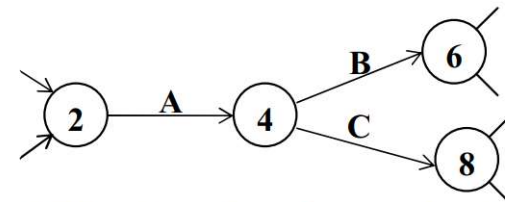
Activity B depends upon the completion of Activity A

(b) Dependent activities



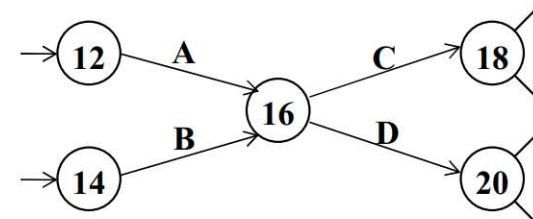
Activity C depends upon the completion of both Activities A & B

(c) A merge



Activity B & C both depend upon the completion of Activity A

(d) A burst

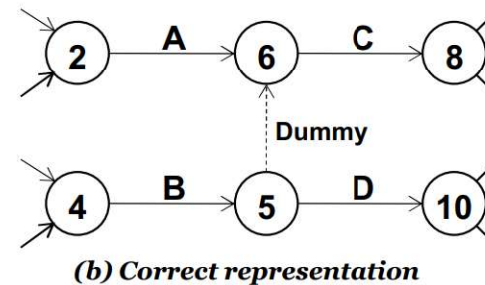
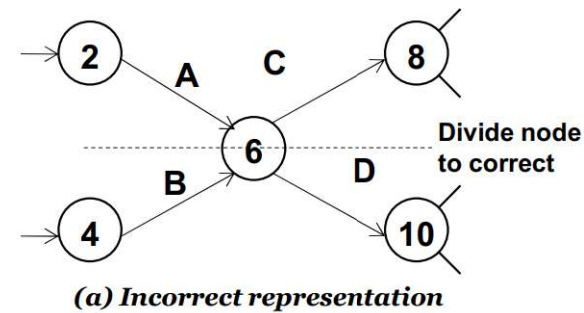


Activities C & D both depend upon the completion of Activities A & B

(e) A cross

ACTIVITY ON ARROW – USE OF DUMMIES

*Activity C depends upon Activities A and B.
Activity D depends upon Activity B only.*



ACTIVITY ON ARROW – USE OF DUMMIES (CONT'D)

Original diagram

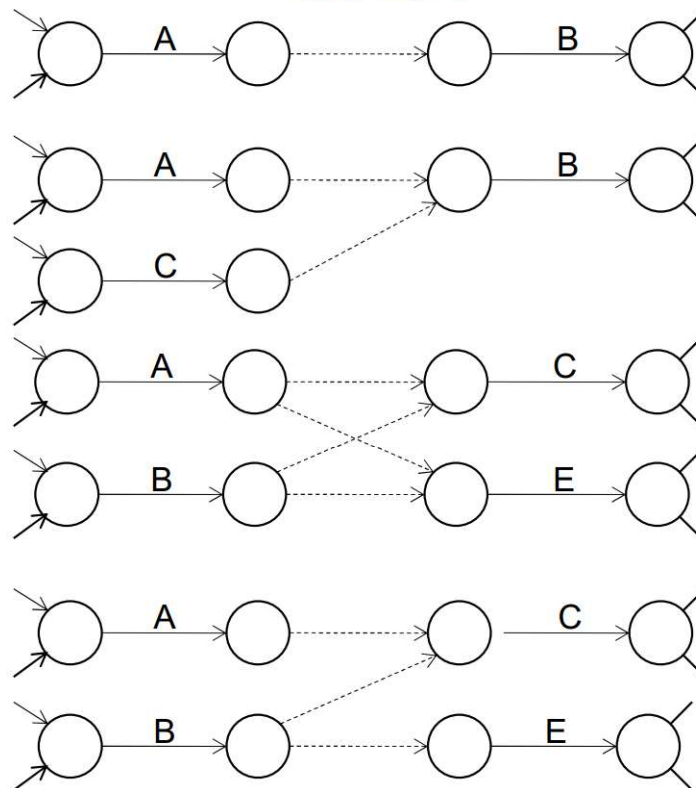
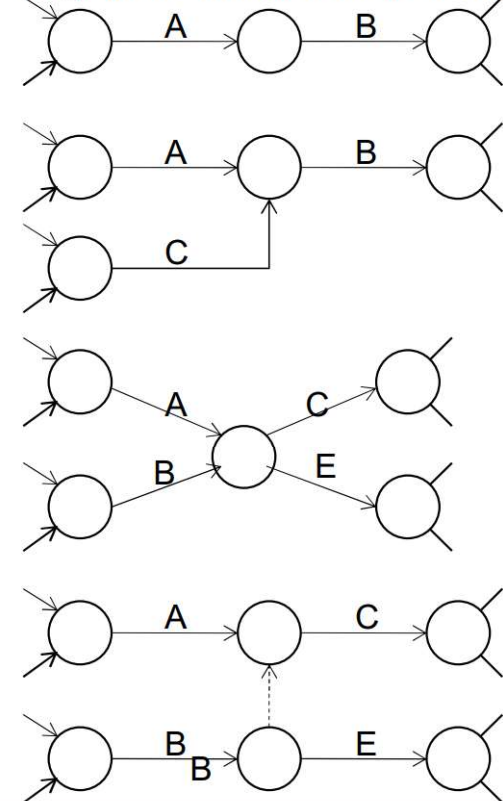
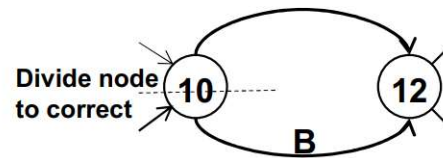


Diagram After removal of redundant dummies

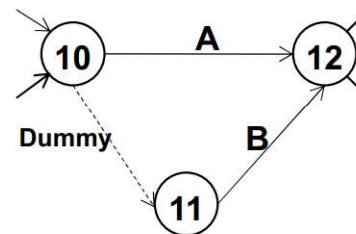


IV

ACTIVITY ON ARROW – REMOVAL OF REDUNDANT DUMMIES



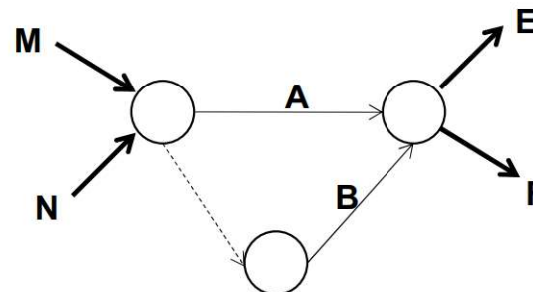
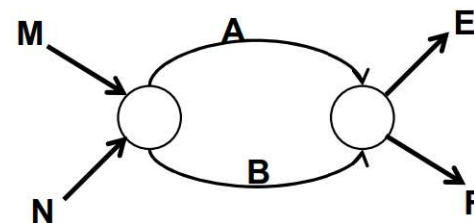
(a) Incorrect representation



(b) Correct representation

ACTIVITY ON ARROW – EXAMPLE

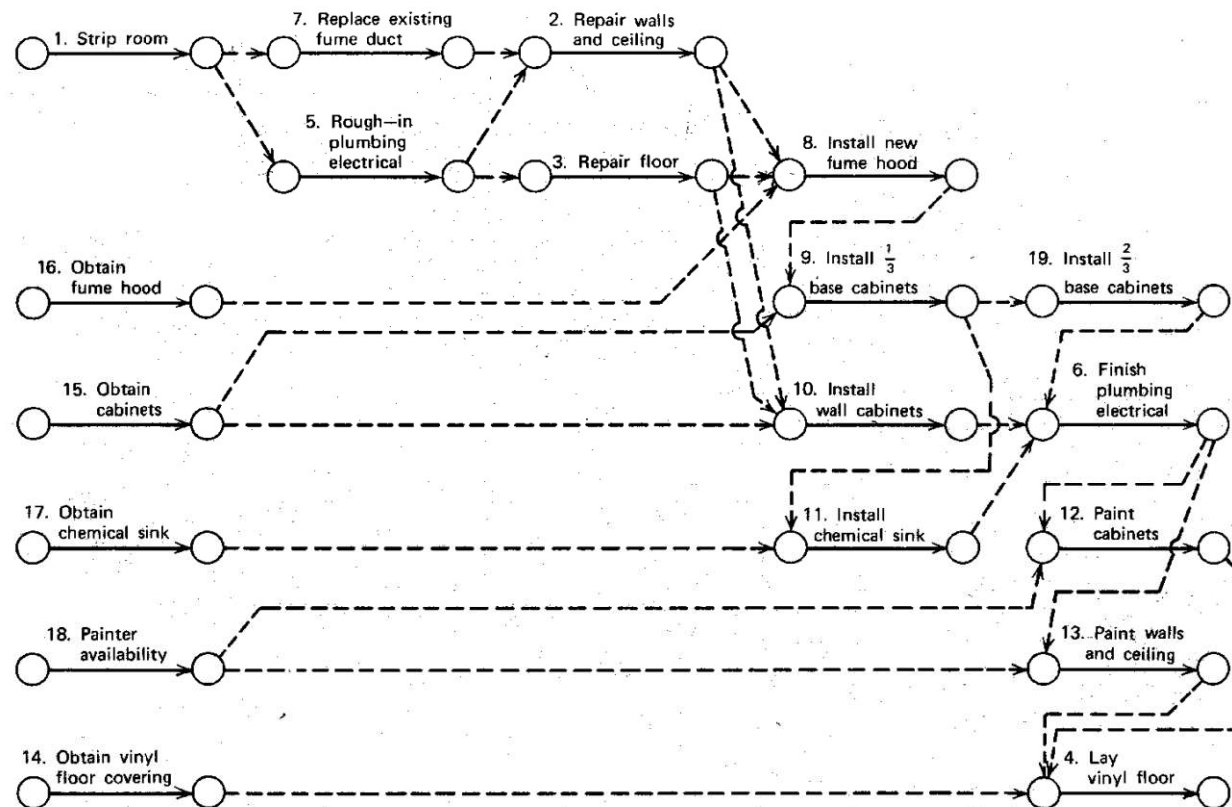
Activity	Depends upon
E	A, B
F	A, B
A	M, N
B	M, N



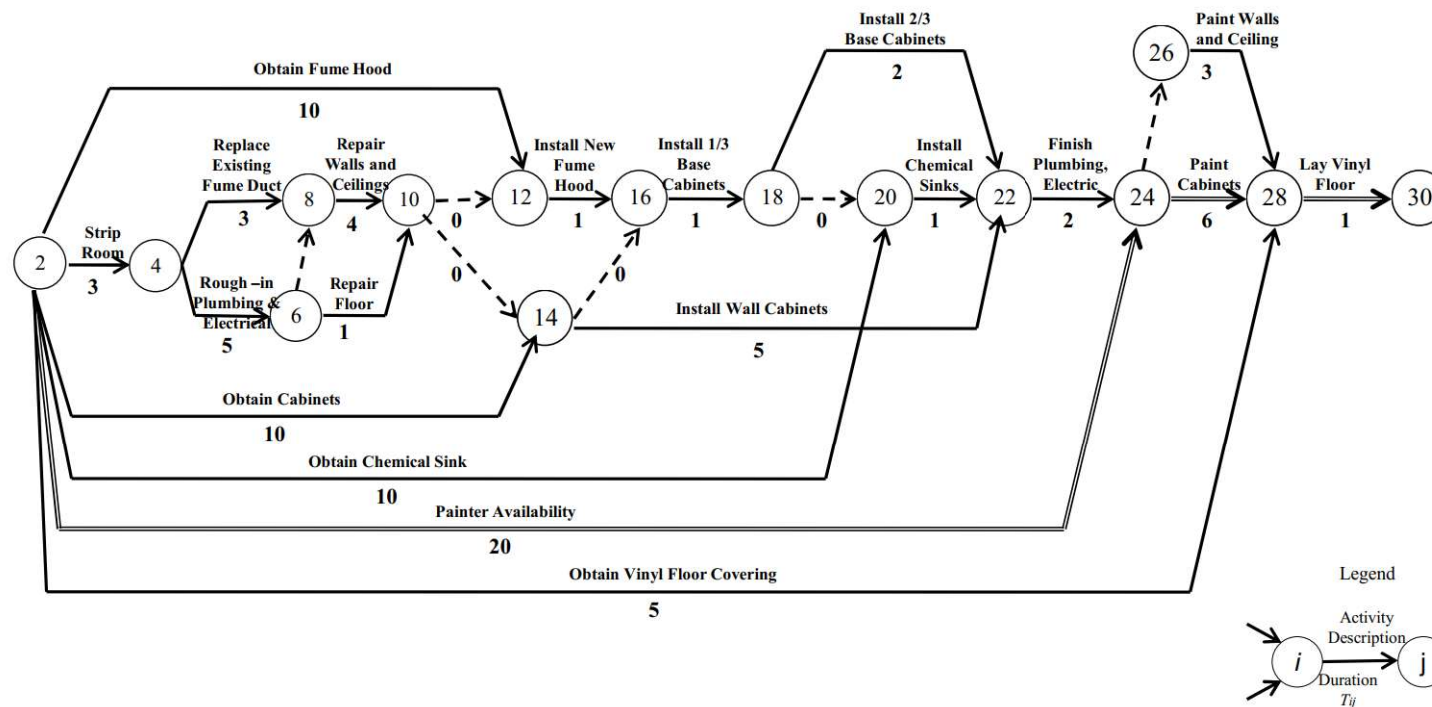
ACTIVITY LIST WITH DEPENDENCIES

No	Activity	Predecessors	Predecessors without Redundancies
1	Strip Room		
2	Repair Walls and Ceilings	1,5,7	1,5,7
3	Repair Floor	1,5	1,5
4	Lay Vinyl Floor	3,12,13,14	3,12,13,14
5	Rough -in Plumbing and Electrical	1	1
6	Finish Plumbing and Electrical	2,3,5,9,10,11,19	2,3,5,9,10,11,19
7	Replace Existing Fume Duct	1	1
8	Install New Fume Hood	2,3,16	2,3,16
9	Install 1/3 Base Cabinets	2,3,8,15	2,3,8,15
10	Install Wall Cabinets	2,3,7,15	2,3,7,15
11	Install Chemical Sink	2,3,5,9,17	2,3,5,9,17
12	Paint Cabinets	6,8,9,10,11,18	6,8,9,10,11,18
13	Paint Walls and Ceiling	2,3,6,8,9,10,18	2,3,6,8,9,10,18
14	Obtain Vinyl Floor Covering		—
15	Obtain Cabinets		—
16	Obtain Fume Hood		—
17	Obtain Chemical Sink		—
18	Painter Availability		—
19	Install 2/3 Base Cabinets	2,3,9,15	2,3,9,15

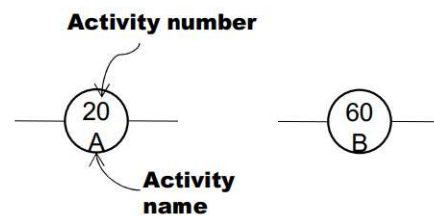
ACTIVITY ON ARROW NETWORK WITH REDUNDANCIES



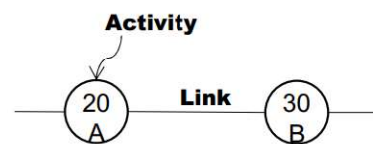
ACTIVITY ON ARROW NETWORK WITHOUT REDUNDANCIES



PRECEDENCE DIAGRAM (AON) – BASIC LOGIC PATTERN

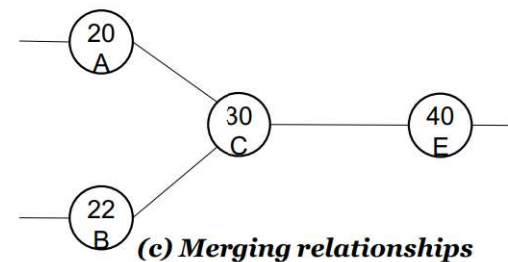


(a) Independent relationships

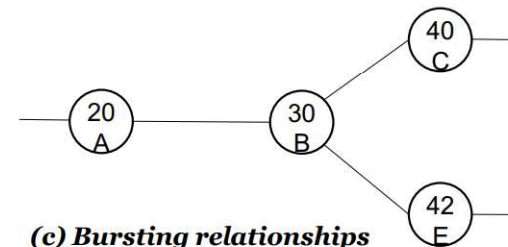


Activity B depends upon the completion of Activity A

(b) Dependent relationships



(c) Merging relationships



(c) Bursting relationships

Basic logic patterns for precedence diagram

AON VS. AOA

- The main difference between AOA & AON is:
 - AOA diagrams emphasize the milestones (events);
 - AON networks emphasize the tasks.

Duration Estimation & Resource Assignment

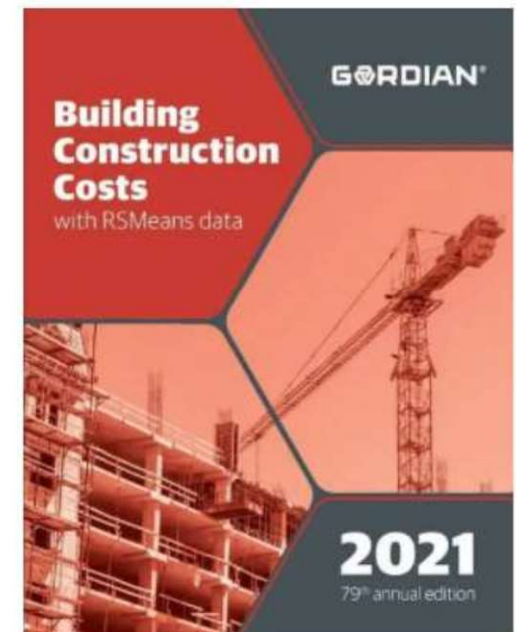
Network Generation

ESTIMATING DURATIONS

- Assigning Resources
- Estimating Durations of Activities
- Knowing how long it takes to complete a task in calendar days as well as the amount of work (level of effort) involved is key to developing an accurate work plan.
- Unfortunately estimating is not an exact science and until hard data is gathered (actual hours applied and actual duration) you will have to rely on the team's experience to estimate task duration and effort.

RESOURCE ASSIGNMENT

- Evaluate each Work Package identified in the WBS, independently of all others
- Select a method & apply a normal level of personnel/labor and required tools and material
- Resource Assignments determine task and project costs



DETERMINATION OF ACTIVITY TIMES

V

- Duration of a task depends on:
 1. Crew selection and Crew productivity
 2. Quantities of Work (assumed constant)

$$\text{Duration} = \text{Quantity of Work} / \text{Crew Productivity}$$

ASSIGNING RESOURCES – CHEMICAL LAB EXAMPLE

Activity List	Duration	Resource						
		Labor	Masons	Sk. Labor	Plumbers	Carpenters	Painters	PM Assistant
Strip Room	3	3						
Repair Walls and Ceilings	4		2					
Repair Floor	1	2						
Lay Vinyl Floor	1			2				
Rough –in Plumbing and Electrical	5	1			1			
Finish Plumbing and Electrical	2				1			
Replace Existing Fume Duct	3							
Install New Fume Hood	1			1				
Install Base Cabinets	3	2				1		
Install Wall Cabinets	5	2				1		
Install Chemical Sinks	1				1			
Paint Cabinets	6	1					2	
Paint Walls and Ceiling	3	1					2	
Obtain Vinyl Floor Covering	5							1
Obtain Cabinets	10							1
Obtain Fume Hood	10							1

V

ESTIMATING ACTIVITY TIMES

- Have the person who will actually do the work provide an estimate for the duration
- Functional managers would typically have at their disposal a large volume of historical data available from which they will derive their estimates

PROBABILISTIC ACTIVITY TIMES

- Often used in R&D or just development activities where tasks are non-repetitive.
- Functional managers would submit their estimates using three possible completion assumptions:
 - Most optimistic Completion time (a)
 - Most pessimistic completion time (b)
 - Most likely completion time (m)

PROBABILISTIC ACTIVITY TIMES (CONT'D)

V

$$\text{Expected time} = \frac{\text{Best} + 4(\text{Most Likely}) + \text{Worst}}{6}$$

$$\text{Standard Deviation} = \frac{\text{Worst} - \text{Best}}{6}$$

EFFORT VS. DURATION

- Effort

- Actual Work Effort
 - Uninterrupted time
 - Actual Time Performing Task
- Units of person-hours, person-days

- Duration

- Working Days (Elapsed Time)
- Fixed Duration Tasks
 - Length of time required to complete
 - Cannot be compressed
- Variable Duration Tasks
 - Duration proportional to resources applied
 - Addition of resources reduces duration (to a point)

$$\text{Duration} = \frac{\text{Effort (Person-days)}}{\text{\# of People Assigned}}$$

EFFORT VS. DURATION

- After creating the WBS, estimate the effort to complete each task.
- For most tasks the amount of effort will determine the duration.
 - For example, say a particular tasks require 8 persons-days of effort. If you assign 4 people to the task the duration will be 2 days. If you assign 8 the duration will be 1 day
- When initially estimating effort, it is easier to assume one person on the task. In this way the effort will equal the duration.

Thank you for your attention!

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